

# SMART INDIA HACKATHON 2026



## Project Report

|                                |                                                                                                                     |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Problem Statement ID</b>    | SIH26084                                                                                                            |
| <b>Problem Statement Title</b> | Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)                                           |
| <b>Theme</b>                   | Disaster Management                                                                                                 |
| <b>PS Category</b>             | Software                                                                                                            |
| <b>Team Name</b>               | TRIKAAL                                                                                                             |
| <b>Team ID</b>                 | 137710                                                                                                              |
| <b>Live Prototype</b>          | <a href="https://vayugati-nowcast.vercel.app/">https://vayugati-nowcast.vercel.app/</a>                             |
| <b>GitHub Repository</b>       | <a href="https://github.com/mannbutani0306/vayugati-nowcast">https://github.com/mannbutani0306/vayugati-nowcast</a> |

## ABSTRACT

Thunderstorms, hail, downburst winds and cloudbursts are among India's deadliest natural hazards; NCRB recorded 2,825 lightning deaths in 2024 alone, 35.7% of all natural-force deaths [1]. Numerical Weather Prediction (NWP) resolves roughly 11–25 km, yet these storms live at 1–10 km and evolve in 15–180 minutes, so warnings arrive late, unlocalised and without an explanation.

**VayuGati Nowcast** is a software-only, CPU-only 0–6 hour convective nowcasting and alerting platform. It fuses live Open-Meteo NWP, archived MOSDAC INSAT-3DR thermal-infrared scenes and NASA GPM IMERG V07 precipitation onto a common grid, tracks storm motion with Farneback dense optical flow, evaluates five hazard heads (hail, downburst, cloudburst, lightning density, convective initiation), and classifies severity as INFO / WATCH / WARNING / SEVERE using Gradient Boosting with SHAP explanations. Every alert follows a database-enforced **DRAFT** → **REVIEW** → **APPROVED** Common Alerting Protocol (CAP) workflow with an immutable audit log, so **VayuGati recommends; the forecaster decides**.

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## 1. PROBLEM UNDERSTANDING

- **Too coarse and slow:** NWP grids of 11–25 km cannot resolve 1–10 km cells that develop and decay within minutes; the gap between two NWP updates is a blind spot.
- **Scattered data:** Doppler radar, INSAT imagery, lightning and NWP sit in separate systems with no common grid or schema.
- **Black-box warnings:** forecasters cannot see why a cell was flagged, which erodes trust and slows decisions.
- **Alert gaps:** warnings are not localised, not multilingual, and there is no reviewed, auditable path from model output to public alert.

The reference event is the 5 August 2010 Leh cloudburst, which produced 4–8 cm of rain in about 3 hours and killed nearly 200 people [2]; VayuGati is built and verified against this class of event.

## 2. PROPOSED SOLUTION : VAYUGATI-NOWCAST

VayuGati is organised as a three-stage pipeline (Fig. 1) that ends in a human decision rather than an automatic broadcast.

- **Ingest & downscale:** Open-Meteo CAPE, CIN and wind guidance, satellite, radar and lightning adapters all feed one schema; fields are bilinearly interpolated to a 1–3 km display grid and labelled as interpolated.
- **AI nowcast:** Farneback optical flow advects reflectivity pixel-by-pixel; five hazard heads and a Gradient Boosting severity model score every tracked cell; SHAP explains each score.
- **Review & publish:** an officer reviews the cell, forecast cone and explanation card, then approves, rejects or cancels with a recorded reason; only approved CAP alerts reach the citizen portal.

### 2.1 Why VAYUGATI-NOWCAST Wins? — Unique Value Propositions

| Differentiator              | Why it matters                                                                                                               |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Proven, not claimed         | Scored against persistence with CSI, POD, FAR and FSS; the speed of a test storm is recovered to within 1 km/h.              |
| Anchored to a real disaster | Replayed on NASA GPM IMERG archives of the 2010 and 2011 Leh cloudburst episodes.                                            |
| Explainable                 | Real SHAP attributions (reflectivity, CAPE, shear, cloud-top temperature, lightning rate, PWAT) on every severity call.      |
| Governed                    | Approval is enforced in the database with row-level security and an immutable audit log, not only in the UI.                 |
| Honestly labelled           | Every feed carries a LIVE / CACHED / OFFLINE / SIMULATED badge; missing data shows “awaiting data”, never an invented value. |
| Inclusive                   | Citizen portal in English, Hindi and Marathi with persona-based advice, plus the VayuGati Saarthi AI guide.                  |
| Low-cost and deployable     | No new radar, sensor or IoT hardware; runs on one CPU core (256×256 grid in ≈29 ms, 1024×1024 in ≈438 ms).                   |

## 2.2. DATA SOURCES

Each source is status-labelled in the interface and on the public Data Sources page.

| Source                       | Use in VayuGati                                                                                | Status                                       |
|------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------|
| Open-Meteo NWP               | CAPE, CIN, lifted index, precipitable water, 0–6 km wind shear for the current atmosphere      | LIVE                                         |
| MOSDAC INSAT-3DR TIR1        | Thermal-infrared cloud-top temperature scenes (GeoTIFF) for initiation and cooling-rate cues   | Archived scenes                              |
| NASA GPM IMERG Final Run V07 | Half-hourly precipitation for Leh 5 Aug 2010 and 25 Jul 2011; monthly climatology via Giovanni | Archived, replayable                         |
| IMD Doppler Weather Radar    | Reflectivity and velocity adapter built to the common schema                                   | Adapter-ready; pilot after IMD authorisation |
| Lightning detection network  | Point-to-grid lightning density adapter (15-minute window)                                     | Adapter-ready / proxy                        |

## 2.3. FEATURES & FUNCTIONALITY

| Module                      | What it does                                                                                                                                                   |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dense optical-flow tracking | OpenCV Farneback motion field from consecutive reflectivity grids; forecast cones and cell ETA for 0–60 min, blended with NWP guidance to 6 h.                 |
| Hail head                   | Logistic probability from the 55 dBZ hail-core signature combined with CAPE.                                                                                   |
| Downburst head              | Gust proxy (km/h) from reflectivity gradients and optical-flow divergence.                                                                                     |
| Cloudburst head             | Marshall–Palmer rain rate ( $Z = 200R^{1.6}$ ) [3]; flags persistence of $\geq 100$ mm/h over four 15-minute frames, plus a monthly-climatology anomaly check. |
| Lightning-density head      | Converts lightning points to a per-km <sup>2</sup> grid density over a rolling window.                                                                         |
| Convective-initiation head  | Detects first crossing of 35 dBZ together with cloud-top cooling of $-4$ °C per 15 min or faster.                                                              |
| Severity + SHAP             | Gradient Boosting [4] classifies INFO/WATCH/WARNING/SEVERE from six features; SHAP [5] shows each feature's contribution.                                      |
| Human-in-the-loop CAP       | CAP-compliant alerts with urgency, severity, certainty, polygon, ETA, English/Hindi/Marathi text; approve, reject or cancel with reason [6].                   |
| Verification engine         | CSI, POD, FAR, FSS [7], bias and reliability diagrams versus persistence, plus IMERG case replay.                                                              |
| VayuGati Saarthi            | AI chat guide that explains approved alerts and portal features in the user's language.                                                                        |
| Accessibility and offline   | Font-scale and high-contrast modes; PWA offline shell for low-connectivity users.                                                                              |

### 3. WEBSITE FLOW

The platform has three public pages and three role-based portals protected by Supabase authentication and row-level security.

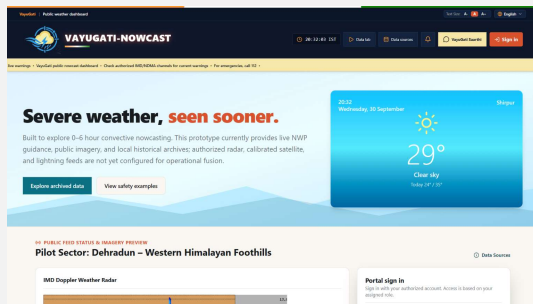
1. **Landing page:** live local weather panel, archived examples and safety guidance, with sign-in.
2. **Data Sources and Data Lab (public):** provenance of every feed and IMERG replay of the Leh cases, without login.
3. **Officer Dashboard:** Leaflet GIS map with switchable layers (radar dBZ, INSAT cloud-top, real INSAT scene, lightning heatmap, optical-flow cones, warning polygons), live ETA countdown per cell, SHAP card, verification tab and the alert review queue (Drafts, Under review, Severe, Approved).
4. **Admin Portal:** district hazard overview, institutional scenarios, response planning and user management, with audit-log status.
5. **Citizen Portal:** approved alerts for the user's area in English, Hindi or Marathi with advice tailored to general citizens, outdoor workers/farmers, commuters and school parents.

#### 3.1. TECHNICAL ARCHITECTURE & TECH STACK

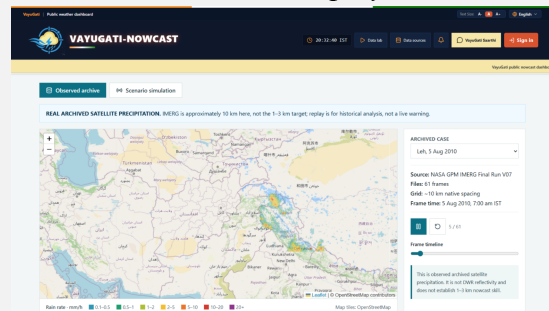
| Layer             | Technologies                                                                                                                                                                             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend          | React 19, Vite, TypeScript/JSX, Tailwind CSS, Leaflet maps, React Router; PWA service worker and manifest                                                                                |
| Backend           | Python, FastAPI with Uvicorn, Pydantic; REST endpoints for ingestion, hazard heads, optical flow, severity prediction, explanation, verification and the assistant                       |
| AI / ML core      | OpenCV (Farneback dense optical flow), scikit-learn GradientBoostingClassifier, SHAP, NumPy, pandas, joblib                                                                              |
| Data handling     | h5py (IMERG HDF5), rasterio (MOSDAC GeoTIFF), netCDF4, httpx (Open-Meteo)                                                                                                                |
| Database and auth | Supabase Postgres with PostGIS geometry; tables for users, regions, weather stations, convective cells, CAP alerts and audit logs; RLS policies and realtime channel on convective cells |
| Deployment        | Docker Compose (backend and frontend), Vercel-hosted frontend, one-command local start (npm run dev)                                                                                     |

## 4. DASHBOARD & FUNCTIONING SCREENSHOTS

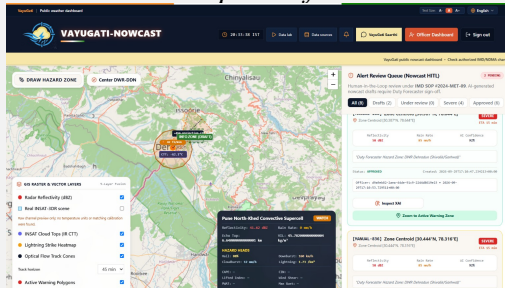
### Home Page *Severe weather, seen sooner*



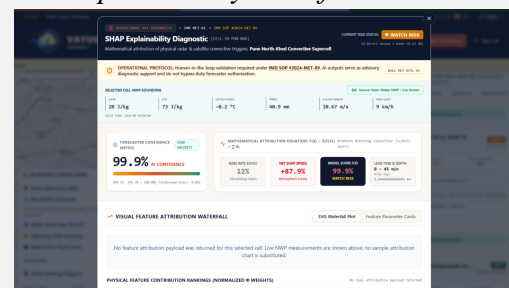
### Data-Lab *Leh 2010 replay*



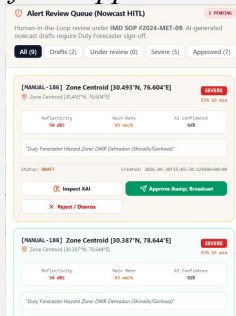
### Officer Dashboard *GIS hazard map with forecast cones*



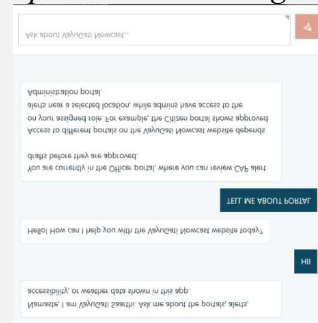
### Cell Analysis Panel *SHAP explainability card for a selected cell*



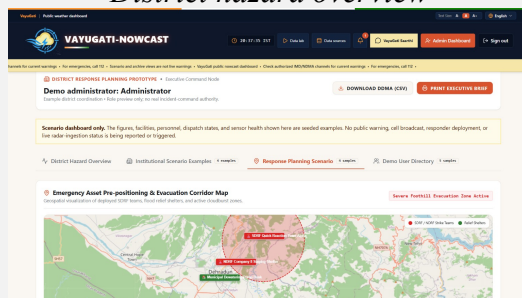
### Alert Review Queue *(Draft → Approve / Reject)*



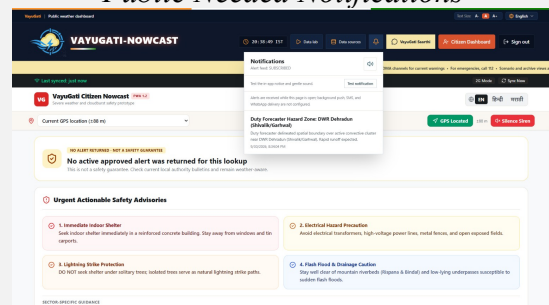
### VayuGati-Saarthi Chat *Gemini-powered scheduling assistant.*



### Admin Portal *District hazard overview*



### Citizen Portal *Public Needed Notifications*



## 5. VERIFICATION & PERFORMANCE

VayuGati is scored against a persistence baseline using CSI, POD, FAR and FSS. In the controlled test-storm benchmark (six storm types, 20 runs each) the nowcast keeps a clear advantage at every lead time, and the 60-minute CSI is 0.755 versus 0.241 for persistence.

| Lead time | Nowcast CSI | Persistence CSI | Note                                |
|-----------|-------------|-----------------|-------------------------------------|
| 15 min    | 0.910       | 0.629           | Test-storm benchmark, all-case mean |
| 30 min    | 0.845       | 0.438           | Test-storm benchmark, all-case mean |
| 60 min    | 0.755       | 0.241           | Test-storm benchmark, all-case mean |
| 180 min   | 0.610       | 0.085           | Test-storm benchmark, all-case mean |

On archived NASA IMERG fields for the Leh 2010 event ( $\geq 1$  mm/h, 30 min lead) the nowcast scores CSI 0.553, POD 0.705, FSS 0.942 and bias 1.002. Because IMERG is a coarse ( $\sim 10$  km), morphing-based product, its skill is close to persistence; it is used to verify data integrity and replay real events, and radar-based verification is planned once IMD feeds are authorised.

## 6. FEASIBILITY, IMPACT & CONCLUSION

**Feasibility.** The stack is open-source and CPU-only, needs no new hardware, and works as an advisory overlay: forecasters review every alert and IMD field protocols are unchanged.

**Impact.** Faster, explainable, human-approved and multilingual alerts give farmers, outdoor workers, district authorities, aviation, transport and emergency teams an additional action window of about 30 minutes in one unified hazard view covering five hazards.

**Governance.** Every alert stores its reviewer, reason and timestamp, exportable for audit.

VayuGati Nowcast turns fragmented weather feeds into a single, transparent and governed workflow, and delivers hyper-local forecasts for safer tomorrows.

## 7. PROJECT RESOURCES

- **Live Prototype:** <https://vayugati-nowcast.vercel.app/>
- **GitHub Repository:** <https://github.com/mannbutani0306/vayugati-nowcast>
- **Prototype Demonstration Video:** <https://www.youtube.com/watch?v=LyKIljgVgBU>

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